

BINF 4070 / COMS 4995 W008
The Future of Personal Health Assistant
Fall 2026

Personal Health Data Foundations

+ Week 01

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AI @ VP&S Initiative



VAGELOS COLLEGE OF
PHYSICIANS AND SURGEONS

COLLABORATORY
AT COLUMBIA UNIVERSITY



Today's Outline

- 1. The New Era of AI for Daily Health and Well-Being**
- 2. Course Structure Overview**
- 3. Everyday Health Data: When Tracking Becomes Continuous**
- 4. Important Considerations for Health Data**

The New Era of AI for Daily Health and Well-Being

When sci-fi becomes reality...

A Personal Health Companion, Imagined in 2014



"Hello. I am Baymax, your personal healthcare companion."

— Baymax, *Big Hero 6* (2014)

AI Agents Span the Health Ecosystem

Patients & consumers

Personal health management · prevention · health information · care navigation · longitudinal monitoring

Clinicians

Documentation · chart summarization · information retrieval · decision support · follow-up

Hospitals & health systems

Scheduling · administrative workflows · coding and billing · care coordination

Researchers & life sciences

Evidence synthesis · cohort identification · trial workflows · biomedical research

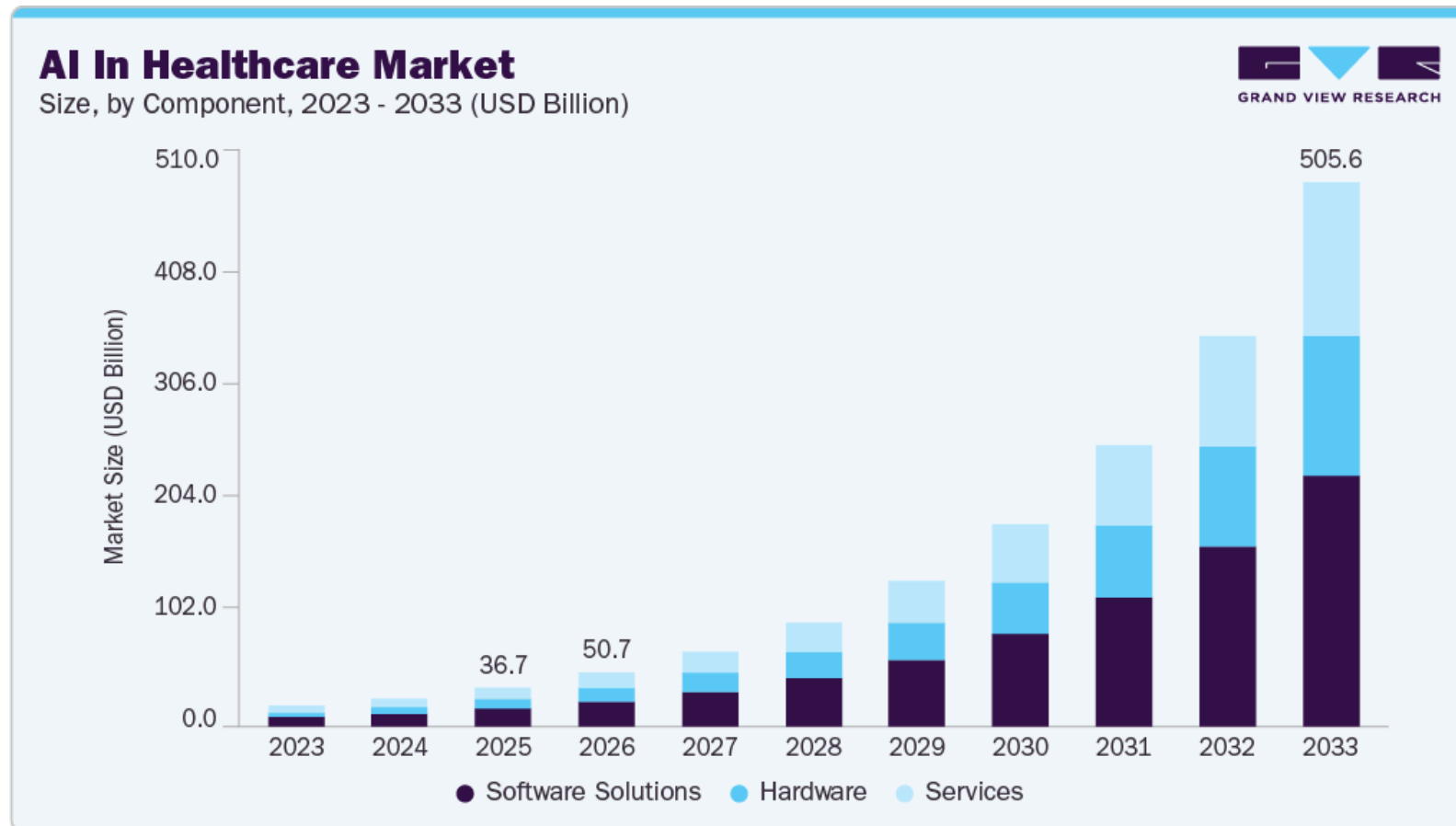
Personal health assistant are one part of a broader ecosystem of agents serving different stakeholders, data, and workflows.

Health Agents Already Support Distinct Jobs

Example and status	Who it helps → what it does
OpenEvidence [®] <i>Commercial medical-information platform · available to verified U.S. clinicians</i>	Clinicians → answers clinical questions with cited medical evidence
ABRIDGE <i>Commercial ambient-documentation platform · deployed in health systems</i>	Clinicians → turns care conversations into draft notes for review
Google Research AI co-scientist <i>Research system · trusted-tester access</i>	Biomedical researchers → generates, critiques, and ranks testable hypotheses

Growth in AI for Health

The exponential growth in health data presents significant opportunities for AI-powered solutions to extract actionable insights and support clinical decision-making.



Personal Health Already Reaches Population Scale

91.0%	41.1%	65.0%
U.S. adults who owned a smartphone as of 2025	U.S. adults who used a wearable to track health or activity as of 2024	U.S. adults who were offered online access to medical records by a provider or insurer and opened them at least once in 2024

Personal-health systems now reach population scale across devices and clinical records.

Personal health spans many aspects: daily behavior, physiology, medication, nutrition, mental well-being, clinical records, and chronic disease.

Our Focus on **Personal Health**: Fragmented Data → Useful Action

Key Question: How can an agent responsibly turn a person's fragmented, longitudinal health data into useful action?

Getting the right data into an agent is itself a major challenge:

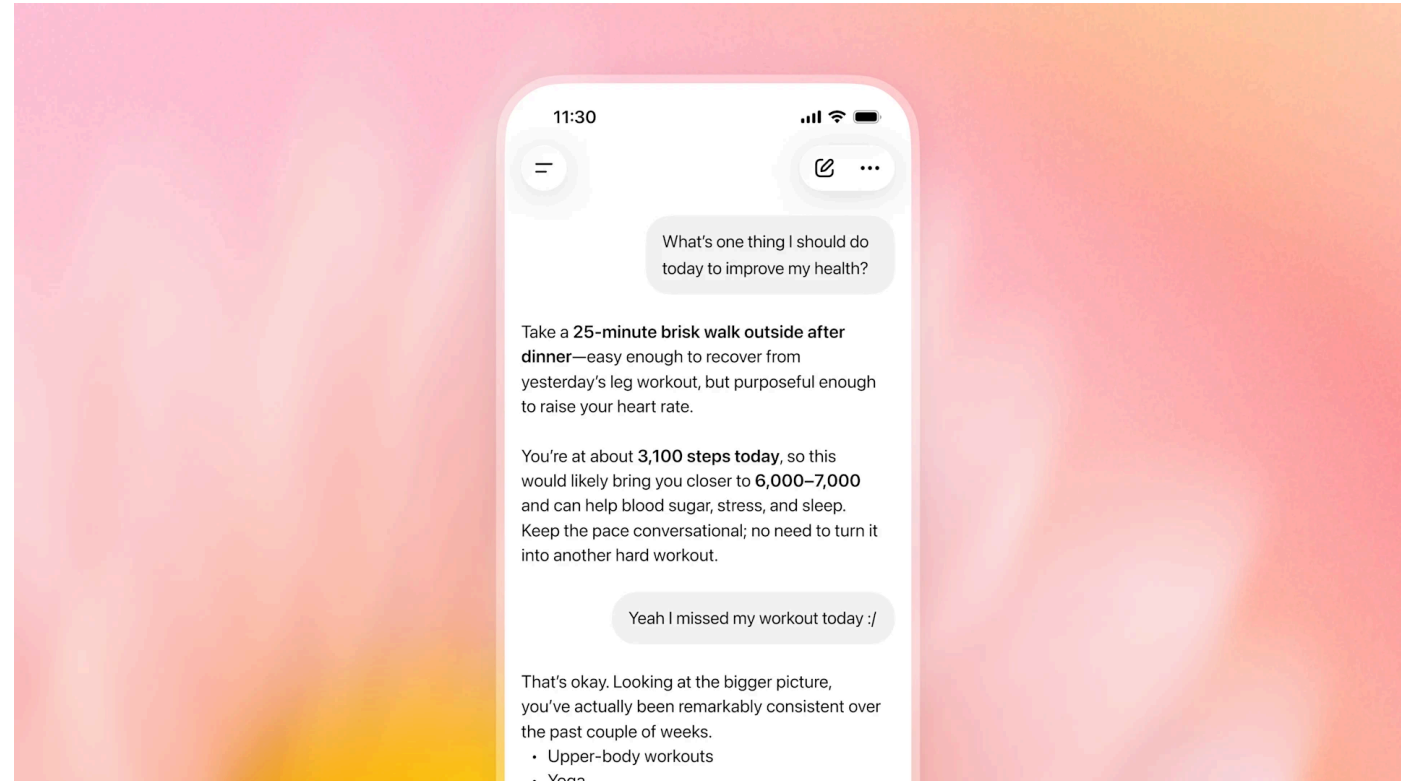
- 59% of U.S. adults reported online health data from more than one type of provider or healthcare organization in 2024
- These secure websites and apps may store and share data differently

Converting these data into actionable insights is another challenge:

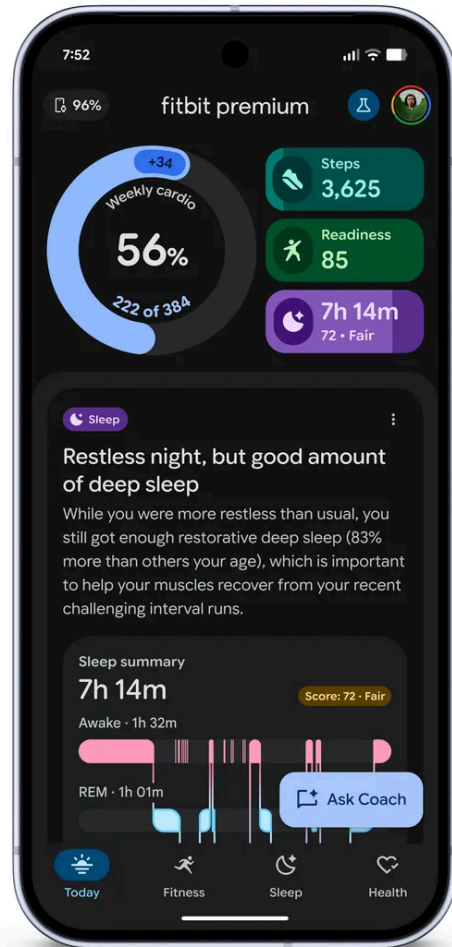
- Interpretation: The same value can mean different things across people, baselines, and situations
- Personalized Action: Guidance must fit the person's goals, explain uncertainty, and recognize when professional care is needed

Health in ChatGPT Can Connect Personal Records

- Health in ChatGPT can now connect supported medical records and Apple Health
- It can compare results over time and help users prepare questions for clinicians



Google Health Coach Uses Longitudinal Data



- Brings goals together with Fitbit and other authorized health data
- Uses conversation for check-ins, plan adjustments, and longitudinal summaries
- Google defines its role as general wellness, not diagnosis or treatment

AI Assistant Capabilities for Personal Health Support

1. **Connected Data** — retrieve authorized personal health and well-being records
2. **Longitudinal Context** — interpret new observations against a personal baseline
3. **Dynamic Interaction** — invite questions, feedback, corrections, and refinement
4. **Bounded Action** — compute, retrieve, draft, or escalate within a defined role and safety boundary

A Personal Health Assistant (PHA)

A **Personal Health Assistant (PHA)** combines authorized personal health data and computational tools to help a person inspect patterns, ask questions, and prepare decisions. Its claims remain traceable to evidence and bounded by intended use.

Many current PHAs use a generative-AI agent, or a team of specialized agents, to connect users with data and tools.

This course uses **assistant** for the user-facing role and **agent** for one possible implementation

Personal Health Agent is another common term, but the broader PHA vision predates LLMs.

A PHA Can Support Multiple Daily Scenarios

Example 1. **Spot a concerning change early**

A PHA notices a sustained change across personal trends and symptoms, prompts timely clinical evaluation under a defined escalation rule, and prepares the supporting evidence.

Example 2. **Support diabetes management day to day**

A PHA connects glucose, meals, medication, and activity to surface recurring patterns, recall clinician-agreed care-plan steps, and prepare questions for the care team.

Note: The assistant supports attention and preparation; diagnosis and treatment decisions remain with qualified experts (e.g., coaches, physicians).

The Role of PHA in Modern Society

“The greatest opportunity offered by AI is not reducing errors or workloads, or even curing cancer: it is the opportunity to restore the precious and time-honored connection and trust—the human touch—between patients and doctors.”

— Eric Topol, *Deep Medicine*, Ch. 1 (2019)

- **Course reading:** Eric Topol, *Deep Medicine* (2019). The full-book PDF is available in CourseWorks.
- **Course-use permission:** Dr. Topol granted permission to use the copy only in our class. Please **do not** share it outside the course.

Our goal is to build a **Personal Health Assistant (PHA)**
whose outputs are **personalized, dynamic, inspectable, safe,**
and aligned with users.

Course Structure Overview

First time offered in Fall 2026. Any feedback would be welcomed!

Planned to be held every fall.

Towards A PHA System

- **Foundations (Weeks 1–3):** understand sources and measurement
- **Data management and grounding (Weeks 5–7):** data store, retrieve, and ground answers
- **Human-centered interactive agents (Weeks 8, 10–11):** design interfaces and agent building
- **Responsibility and integration (Weeks 13–14):** privacy, red-team, integration.

This is a **project-oriented course**:

- Week 4: Capstone Project Pitch & Team Formation
- Week 9: Project Proposal with Low-Fidelity Prototype
- Week 14: Project Demonstration and Showcase - A Dedicated Room where Everyone Will Present Their Demos

Two short quizzes at Week 7 and 11 (close-book). No mid or final exams.

Lecture and Hands-on Lab

This is a **hands-on course**:

- **Time:** Friday 8:30–11:30 AM ET
- **Location:** Mudd 825 (Morningside Campus)
- **Lecture** ~1.5-2 hrs: establish concepts and shared vocabulary. Mostly by Orson Xu and Eugene Wu.
- **Lab** ~1-1.5 hrs: apply them to a working artifact
- **Attendance:** submit at the end of the lab session for attendance, incomplete is fine.
- **By next Wednesday:** submit the same notebook after completing the take-home tasks and reflection

A large amount of content may not be covered extensively during the lecture, but learned during practice. So, get your hands dirty!

Instructors



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Computer Science (affiliated)

Research: Human-computer interaction,
human-centered AI, everyday sensing, health
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Eugene Wu

Associate Professor
Computer Science

Research: Data systems, human-data
interaction, reliable AI agents

Office: Morningside Campus, Mudd Building,
Room 421

Teaching Assistants



Millie Wu

CS PhD student

Office hours: Monday 6 - 7 PM ET

Location: CS Lounge



Reya Vir

CS PhD student

Office hours: Wednesday 10 - 11 AM ET

Location: CS Lounge

Course Materials

- Main Course Website: <https://personal-health-agent.github.io/>
 - Course syllabus, project description, weekly slides, most reading materials
- Github Links: <https://github.com/personal-health-agent/binf4070-2026>
 - Weekly slides
 - Python notebooks for Lab sessions
 - You will use Colab - More to elaborate in the lab session
- Course Works: <https://courseworks2.columbia.edu/courses/251854>
 - Assignments (Labs, quizzes, project), grades, announcements, and Ed Discussion forum
 - Some copyright-protected reading materials

Build with Your Own Data

- Each of you will have the option to get a **Fitbit Air** (wk 3).
 - If you get one, you agree that your final project will include Fitbit as part of the PHA system. Using your own data would be a plus.
 - You will need to sign an electronic form and upload your Student ID card to get the device.
 - You will need to return the device at the end of the course
 - Please try your best to protect the device!
- You are not required to get a device. **Synthetic data** provide a complete path for the labs and project
 - Getting a device does not affect your score; it is not part of the rubric



AI Use Policy

You are **encouraged** to use AI. One of the learning objectives is to learn how to prompt AI effectively.

Each of you will get a **pre-paid API key** (up to ~20M tokens, TBD, wk3). When used up, you are on your own. You are free to use your own API key or other AI tools.

When using AI, disclose the assistance, verify its outputs, and keep identifiable data protected.

You don't want to be replaced by AI in 2030. So I trust you that...

- You **won't skip the class** and imagine that AI can just do everything for you
 - Well, technically you *can*, but then why are you even taking the class? :)
- You **won't blindly throw the lab notebook to AI** and just copy the output
- You **will think creatively and critically** about the problem you want to solve - the humane part
- You **will verify the outputs from AI** and use them as a starting or ending point for your own work

Grading Policy

Component	Weight	Notes
Weekly labs	5% in class (10 x 0.5%) 15% take home (10 x 1.5%)	12–13 sessions; only 10 required.
Two quizzes	20% (2 x 10%)	Week 7 & 11 (on the class)
Proposal + presentation	15%	Week 9 (on the class)
Demo / showcase	15%	Week 14 (on the last class)
Final implementation	15%	Due: Week 15
Writeup + reflection	15%	Due: Week 15

Optional pre–Week 4 capstone project pitch: **+1% extra credit**.

Late policy: Three “grace days” per student/team across the term; no request needed. After that, 10% off per day unless arranged in advance.

Capstone Project

- You will form a team of **3–4**, ideally with diverse backgrounds
 - Our class has a balanced mix of health and CS students
- Each team will choose a specific target user group and question to address with a PHA system
- The system should use at least one of: (1) clinical/EHR, (2) everyday sensors, or (3) self-report source
 - Will introduce these three sources later in the lecture
 - Multiple data modalities are encouraged and preferred
- Build an LLM, RAG, or agent component with a usable interface
- The system can be any reasonable format: website, mobile app, desktop app, command-line tool, or something creative...

Capstone Project

Read more about the capstone project in the course syllabus:

personal-health-agent.github.io/project-details/

Start ideation and team formation early!

- Week 4: Project Pitch & Team Formation
- Week 9: Project Proposal with Low-Fidelity Presentation
- Week 14: Project Demo and Poster; Writeup Due Dec 16

Join the Waitlist

This year we are splitted between DBMI & CS
35 slots in BINF 4070; 15 slots in COMS 4995
W008.

Scan this QR Code to tell us that you joined the
waitlist and are interested.

We received 100+ requests to join the class so we
had to approve a subset.

We will priortize those who attended the class
today. As of Wednesday, we have ~15 open slots.



Questions about the Course?

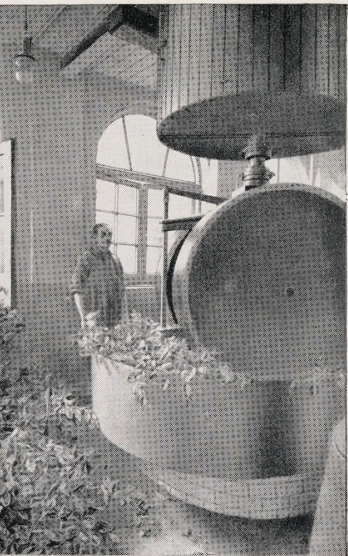
Break

An Overview of Everyday Health Data: When Health Tracking Becomes Continuous

A Historical Perspective: Quantified Self

Quantified Self refers to practices that use personal data to learn about one's health, behavior, or environment. A useful self-tracking practice starts with a question the data can help answer.

Coined in 2007, it's a cultural movement and practice of tracking daily life data, such as sleep, heart rate, steps, and mood, using technology to gain self-knowledge through numbers.



FRESH
BELLADONNA
LEAVES

About to be expressed for juice and for making the green extract. It is extremely important that this be done promptly to avoid fermentation and consequent deterioration of the product. The fresh herb is gathered as soon as the sun is up, and expressed and treated before sunset.

'WELLCOME'
CHEMICAL
WORKS



LOCK
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ulatum)
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(*Conium*
(m)). The
eaves and
are collected
e fruit begins

'WELLCOME'
RIA
CA FARM

a fine loamy soil, varied by sandier uplands. Here the firm have for the last six years been cultivating medicinal plants under the immediate superintendence of pharmaceutical and botanical experts. The farm was established, firstly, to provide opportunities and materials for research and experiment, and, secondly, to supply the manufacturing departments with medicinal herbs of proper quality.

"A visit to the farm shows that the greater part is devoted to the cultivation of staples; but a number of plots are used for experimental crops. Among such are meadow saffron (*Colchicum autumnale*), with its pale-purple flower. Lavender, peppermint, and French roses grow side by side. Senega and the unpretentious taraxacum, with its bright yellow petals, occupy other spaces. Ginseng, the root that plays so important a part in Chinese medicine, is also grown. *Podophyllum peltatum*, *Scopolia atropoides*, *Datura meteloides*, sea poppy (*Glaucum luteum*), and *Grindelia robusta*, are other plants that one does not usually find growing on a scale greater than the experimental; but the plots of *Hydrastis canadensis* are botanically and commercially the most interesting on the farm, in view of the fact that we are coming within measurable distance of the end of the natural supply from North America.

"It is grown at the 'Wellcome' Materia Medica Farm in the open under perfectly natural conditions, in a little woodland dell shaded by tall elms and bramble bushes; and, in another part of the farm, under a lattice-work structure, an effort to re-create the conditions of the native home of golden seal, which is in rich, moist woods from Canada to Carolina. The growth under the latter conditions is more generous. In this case the plants are protected from the noonday heat.

"The purpose which Burroughs Wellcome & Co. had immediately in view when they established this farm, i.e. supplying the products of the field direct to their Works, has been fulfilled, and the farm has in that respect passed the experimental stage, since they have experienced the benefits of conducting a farm in conjunction with the production of pharmaceutical preparations. On the research side, experiment goes on, especially in regard to selection and cultivation of strains which have been found by chemical and physiological tests to be the most active."

A Diary Records Chosen Moments

- The person chooses when and what to record
- Their words preserve context: "pain woke me at 3 a.m."
- Recall, interpretation, and effort shape the entry
- Potential issues:
 - no symptom
 - a forgotten entry
 - being too unwell to record

Wearables Create Longitudinal Streams

- Sensors can collect measures without manual entry, sometimes called "*mobile sensing*" or "*passive sensing*".
 - Physical activity
 - Sleep
 - Heart rate / heart rate variability
 - SpO₂ / oxygen saturation
 - Blood pressure, blood glucose (emerging)
 - ...
- Repeated and longitudinal data make personal baselines and change visible
- Charging, motion, and synchronization create gaps in the data stream





Beyond Consumer Devices: Medical Wearable Devices

- A **continuous glucose monitor (CGM)** can reveal within-day glucose dynamics
- **Insulin pumps** deliver insulin under the skin continuously or on demand
- **Ambulatory ECG monitors** record heart electrical activity during daily life to help evaluate intermittent rhythm problems
- Access depends on cost, indication, comfort, training, and health-system support
- Each device contributes a specific slice of physiology or care

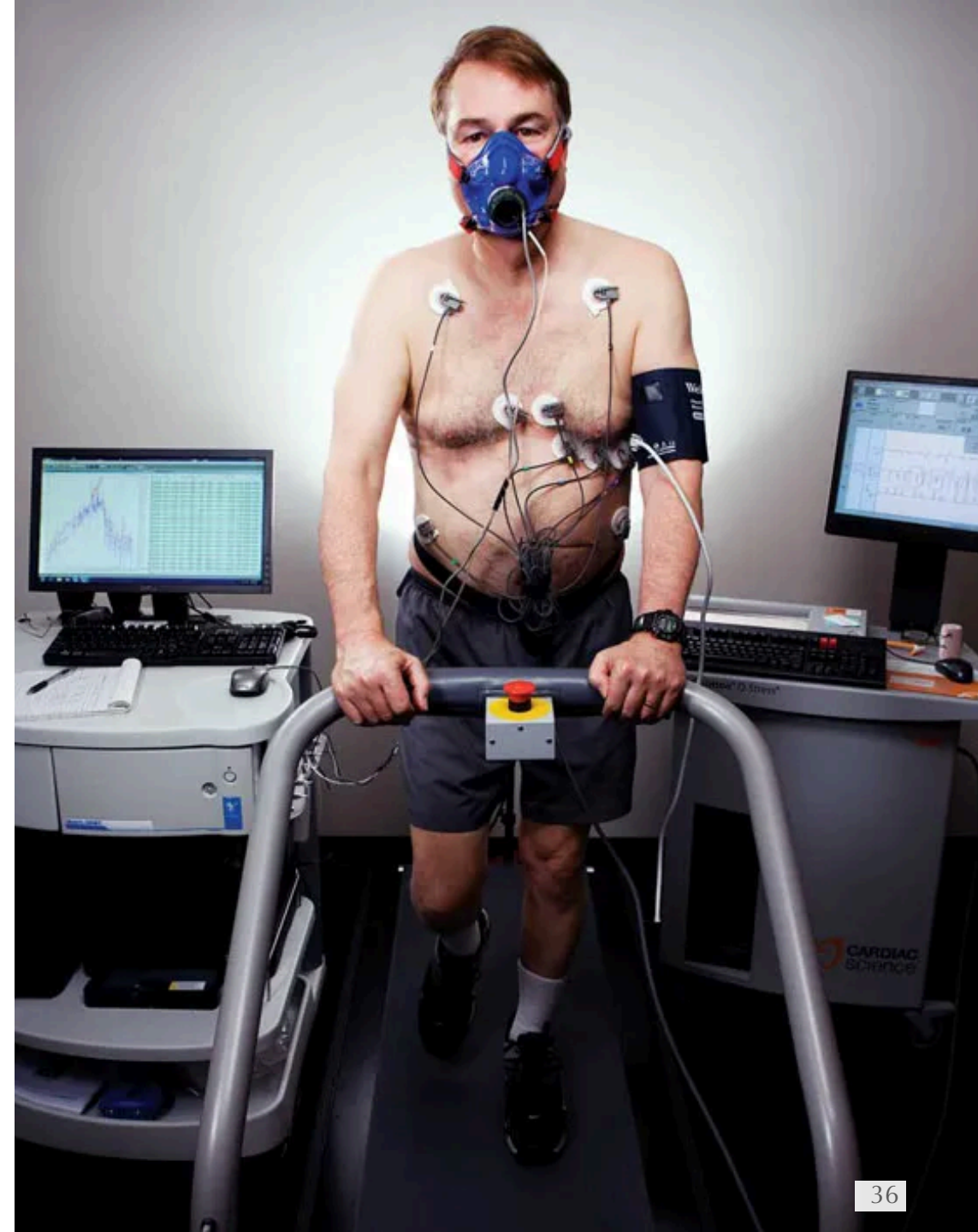
Longitudinal Data Can Prompt Inquiry

Internet pioneer, Larry Smarr, tracked activity, diet, symptoms, and roughly 100 biomarkers.

- Persistent inflammatory markers stood out from his personal baseline.
- Symptoms, specialist judgment, testing, and colonoscopy shaped the diagnosis.

The record prompted clinical investigation.

Colonoscopy and imaging led clinicians to consider **colonic Crohn's disease**— a type of inflammatory bowel disease affecting the colon.



A Data Ecosystem that Joins Distinct Sources

A **personal health data ecosystem** is the set of records and data streams that can describe a person's health. Each source adds a partial view shaped by how and why it was produced.

Three representative modalities organize the evidence:

- **Clinical/Electronic Health Record (EHR)** — care delivery → diagnoses, labs, orders, and notes
- **Everyday Sensors** — Data (e.g., mobile, wearable, connected-device, and network traces) + algorithms → physiological estimates and behavioral features
- **Self-report** — a person's entry → symptoms and lived context

Choose and combine sources around the user's question. No single modality captures the whole picture of the user.

Clinical Records Follow Care

- **Administration:** demographics, insurance and claims, dates, clinicians, and locations...
- **Conditions/history:** diagnoses, problem lists, allergies, immunizations, and family history...
- **Treatment:** medication orders, procedures, referrals, and care plans...
- **Measurements:** vital signs, laboratory and pathology results, imaging reports, and ECG waveforms...
- **Narrative and media:** progress notes, discharge summaries, messages, questionnaires...

EHRs are **encounter-centered**: rich when a person is in a clinic, but often silent between clinical visits.



The screenshot displays a web application for hospital management. On the left is a dark sidebar with a search bar and a menu containing: Inventory, Patients (selected), Patient listing, Assessments, Visits, Procedures, Appointments, Labs, Imaging, Medication, and Users. The main area is titled 'Patients' and features tabs for Patient Listing, Assessments, Visits, and Procedures. A '+ new patient' button is in the top right. Below the tabs is a table with columns: Patient ID, Last Name, First Name, Gender, and DOB. The table lists ten patients, each with a 'Delete' button (red) and 'View' (blue) and 'Edit' (grey) buttons.

Patient ID	Last Name	First Name	Gender	DOB			
11A43873...	Simpson	Bart	M	1/14/1994	Delete	View	Edit
11A43873...	Beardly	Jasper	M	8/5/1932		View	Edit
11A43873...	Bouvier	Ling	F	3/23/2001		View	Edit
11A43873...	Brockman	Kent	M	11/12/1967		View	Edit
11A43873...	Burns	Charles	M	9/2/1944		View	Edit
11A43873...	Skinner	Seymour	M	6/7/1964		View	Edit
11A43873...	Wiggum	Ralph	M	10/8/1993		View	Edit
11A43873...	Szyslak	Moe	M	4/27/1954		View	Edit
11A43873...	Van Houten	Milhouse	M	2/16/1994		View	Edit

Everyday Sensors Produce Behavior Features and Estimates

- **Behavioral feature**

GPS coordinates → location observations → time away from home → possible routine change

- **Physiological estimate**

PPG signal → reflected-light variation → heart rate / heart-rate variability → health monitoring

In both cases, algorithms transform observations into derived features or metrics for later data analysis or insight generation.

Sensor data are **partially observed**. They need to be paired with other evidence before drawing a conclusion.

Self-Report Preserves Lived Experience

- Self-report can be
 - Structured: surveys or ecological momentary assessment (EMA)
 - Free text: diaries
 - Media: food, photo, or video logs
- A person's report gives subjective appraisals (e.g., pain, fatigue, mood, and stress) that a sensor cannot directly capture
 - "Pain woke me at 3 a.m.; Pain level 7/10; Area around my shoulder and knee" places an event in lived context
- Prompts near the moment can reduce recall error -- the benefit of EMA

Self-report is **inherently subjective** and can be shaped by recall, interpretation, burden, and willingness to disclose.

Other Sources Extend the Ecosystem

- **Genomic and molecular assays** add inherited or biological context
- **Environmental and social conditions** change what measurements mean
- **Pharmacy claims and home devices** add other institutional records
- ...

A well-scoped data system names the sources it uses and the relevant context it does not yet capture.

Fragmentation Crosses Systems

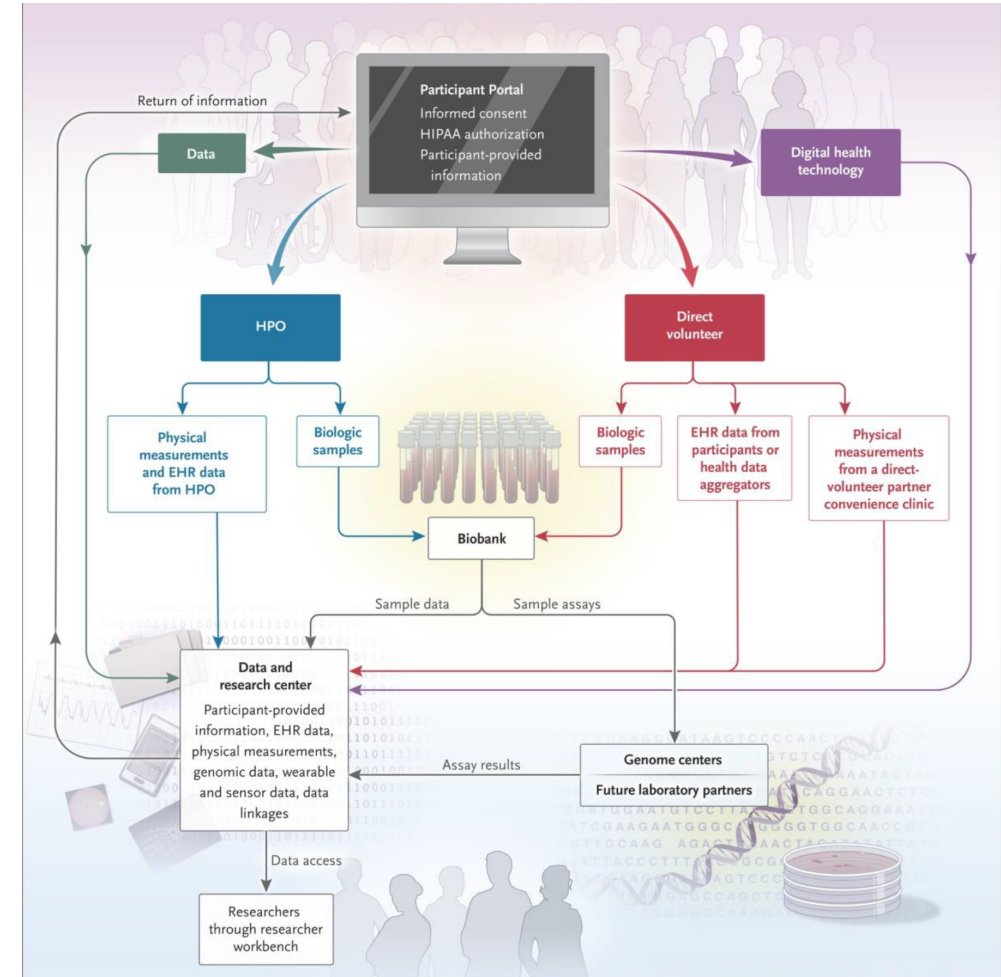
Data fragmentation is the separation of one person’s data across organizations, platforms, formats, identifiers, and access rules.

A hospital, a wearable platform, and a small clinic may identify and structure the same person’s records differently.

Standard	What it helps align
Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM)	Observational data for analysis
Fast Healthcare Interoperability Resources (FHIR)	Resource-based exchange of healthcare information
Open mHealth	Schemas for mobile and wearable health data

Integration Makes Gaps Visible

- Effective integration combines technical alignment, institutional agreements, and governance.
- Integration makes gaps visible and supports cross-source research questions.
- Ongoing effort: The **All of Us Research Program** combines questionnaires, EHRs, digital health technologies, and biospecimens.
 - Coordination supports cross-source research questions
 - Participation, pipelines, and access tiers become explicit system components



Important Considerations for Health Data

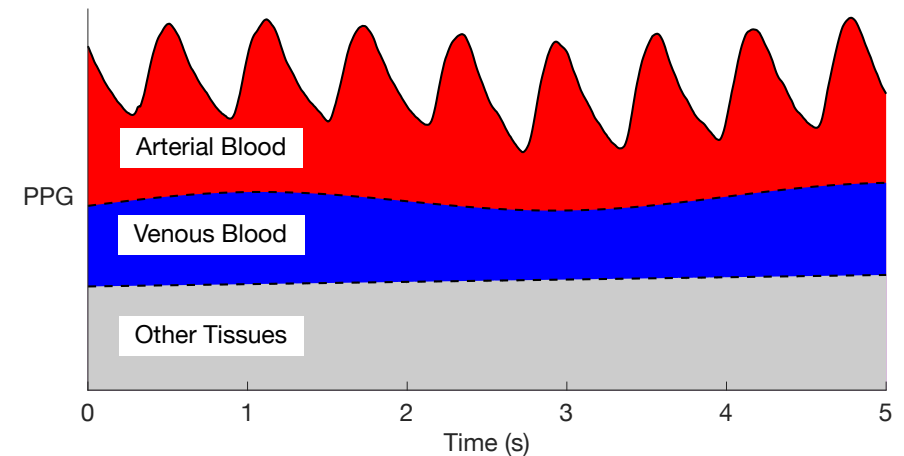
Measurement, missingness, error, inspectability, access, privacy, governance
help determine what an AI health assistant can support.

Measurement Is an Estimate

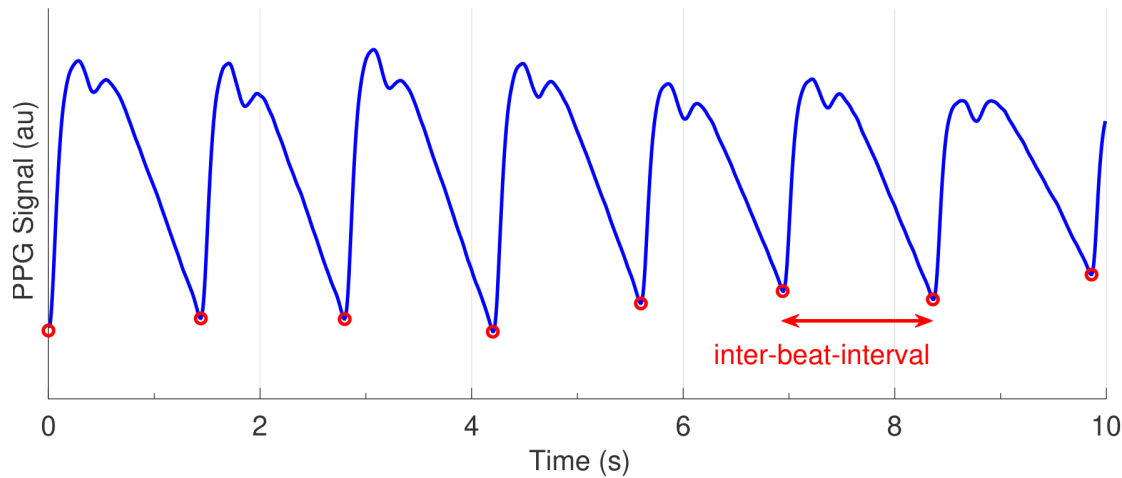
A **derived metric** is estimated from observations by an algorithm. A sensor records a signal; software filters artifacts, detects patterns, and produces the value shown to a user.

Example: Photoplethysmogram (PPG)

- Wrist PPG measures changes in reflected light
- Arterial pulsation is the main contributor; tissue and respiration also shape the signal
- Heart rate emerges from downstream processing
- Uncertainty depends on the device, metric, population, setting, and reference measure



Measurement: Reading Pulse from PPG



- Peaks and valleys can support pulse-interval estimation
- Signal shape depends on sensor, body location, contact, and physiology
- A derived metric carries uncertainty from signal acquisition
- **Motion artifact:** distortion created when a sensor moves relative to the body
 - In wrist PPG, motion artifact can obscure the blood-volume changes used to estimate pulse rate, a leading cause of uncertainty.

Measurement: Evaluate Accuracy in Context

A strong measurement evaluation reports these conditions before generalizing performance:

1. Which **device**?
2. Which **metric**?
3. For which **population**?
4. During which **activity**?
5. Against which **reference measure**?
6. ...

Missingness May Carry Information

Informative missingness occurs when the reason of missing data is related to the person or outcome.

- Some missingness is **non-informative**: a sensor may fail to record because of a technical issue unrelated to the person, e.g., hardware glitch, software crash
- Some missingness is **informative**: illness or a change in routine may lead a person to remove a device; symptoms may prevent a clinic visit or diary entry.
- Informative missingness can bias analysis and reveal aspects of the person's state or context.

Handling **missing data** remains an active research area.

Error Costs Depend on Context

An error's cost depends on **who is affected, the setting, and what decision or action follows:**

- **False reassurance (false negative):** misses a real concern → delayed attention or care
- **False alarm (false positive):** flags a concern that is not present → unnecessary worry, extra cost, or alert fatigue
- **Measurement error:** reports an inaccurate or imprecise value → distorts trends, thresholds, or downstream conclusions
- **Algorithmic bias:** produces systematically different error rates across groups → unequal reliability and potential harm

When evaluating a measure or algorithm, ask:

Who is affected? · In what setting? · What action follows? · Which error is more costly?

Choose metrics and thresholds based on **consequences, not accuracy alone.**

A Strong Answer Shows Its **Inspectable** Basis

An **inspectable answer** shows each claim's source, time window, transformation, uncertainty, and what evidence could change it.

Question: "Why was I so tired this week?"

1. Retrieve relevant sleep, activity, symptom, and medication records.
2. Align the observations in time.
3. Compare plausible explanations and mark uncertainty.
4. Cite supporting, conflicting, and missing evidence; recommend appropriate escalation when professional care is warranted.

Four Layers of an **Inspectable** Answer

- **Observation:** “The record ends the sleep interval at 05:42.” Cite the record.
- **Derived metric:** “Average duration fell 72 min.” Show the method and window.
- **Inference:** “Short sleep may have contributed.” Mark uncertainty.
- **Decision:** The user chooses a next step based on evidence, values, risk, and authority.

Question: “Why was I so tired this week?”

Example answer:

"Your sleep duration averaged **5 h 48 min/night this week**, about **72 min below your previous 4-week average** (*Aug 3–9 vs. Jul 6–Aug 2*). This **may have contributed to your fatigue**, but the data do not establish a cause.

I do not have information about illness, stress, or medication changes. If those factors changed, or the fatigue persists or worsens, the interpretation should be revisited."

Ways to Present an Answer for **Inspectability**

- **Visualization** — dashboards, charts, diagrams, overlays: make magnitude, trend, and comparison legible at a glance
- **Language** — reports, messages, narratives, Q&A: carry the evidence, caveats, and what remains unknown
- **Haptic or audio cues** — vibration, tone, spoken cue: reach the user with no screen and little attention

Combine these forms to fit the user's abilities, device, interaction context, the information being conveyed, and its uncertainty.

E.g., A watch buzz can notify; text or speech can explain what changed and why.

Data Access: Databases

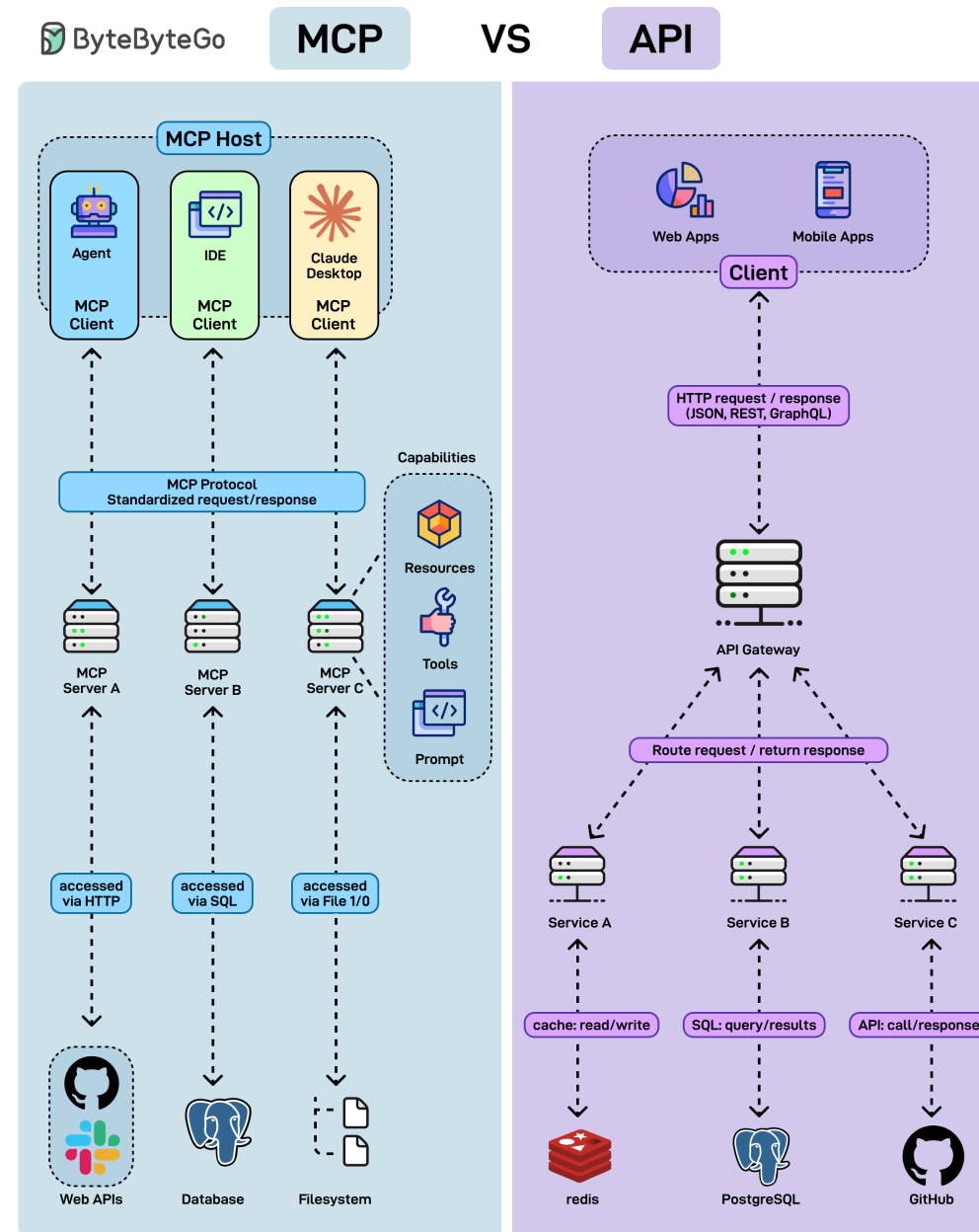
A **database** is an organized collection of data. A database management system uses a **data model**, **schema rules**, and a **query interface** to store and retrieve it.

Pattern	Organizes data as	Possible use
Relational DB	typed rows in tables, joined by keys (SQL)	MIMIC-IV loaded into PostgreSQL
Time-series DB	timestamped points per stream	a heart-rate stream in InfluxDB
Document DB	nested, self-describing records (JSON)	FHIR Observation JSON in MongoDB
Graph DB	nodes and the edges between them	drug-condition links in Neo4j
Vector index	embeddings retrieved by similarity	clinical-note search with pgvector

Patterns overlap; **Weeks 5–7** revisit their tradeoffs.

Data Access: APIs & MCP Serve Different Integration Layers

- An **Application Programming Interface (API)** specifies software requests and responses to a service
- The **Model Context Protocol (MCP)** is a client-server protocol AI hosts use to discover and use server-exposed **resources, tools, and prompts**
- MCP often sits over APIs, databases, or files to make AI agents' life easier
- **Week 2:** FHIR resources and clinical exchange
Weeks 10–11: MCP and permissioned tool use



Data Access: MCP Example

APIs define service-specific operations; MCP gives AI agents a standard way to discover and use server-exposed resources, tools, and prompts.

1. **Request** — “Show my last seven days of sleep.”
2. **Discover** — the PHA AI assistant finds a bounded `get_sleep(start, end)` tool
3. **Validate and call** — the MCP server checks the request and authorization, then invokes the underlying health API
4. **Return** — typed sleep records and provenance support an inspectable answer

MCP is broader than read-only data access: servers can expose **resources, tools, and prompts**, and every action must remain permissioned.

Data Access: API Example

Google Health API v4 represents a recorded or computed metric as a **DataPoint**.

- Typed payloads can represent steps, heart rate, or sleep
- `dataSource` can retain device, app, platform, and recording method
- Provenance documents origin while proprietary algorithm details may remain unavailable
- **Today's lab:** use the same synthetic record shape; live access is optional

The screenshot shows the Google Health API v4 documentation page for the **DataPoint** resource. The left sidebar contains a navigation menu with sections like Overview, v4, errors, REST Resources, and users. The main content area displays the resource name, a description, and its JSON representation. The JSON shows fields for name, dataSource, steps, floors, heartRate, sleep, dailyRestingHeartRate, dailyHeartRateVariability, exercise, weight, and altitude. The right sidebar shows page information and a list of resources on the page.

Google Health API

Guides Reference Support

Search / Language Sign in

Filter

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rollUp

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ActiveEnergyBurnedRollupValue

ActiveMinutesRollupValue

ActiveZoneMinutesRollupValue

ActivityLevelRollupValue

AltitudeRollupValue

BloodGlucoseRollupValue

BodyFatRollupValue

CaloriesInHeartRateZoneRollupValue

Resource: DataPoint

A computed or recorded metric.

JSON representation

```
{
  "name": string,
  "dataSource": {
    object (DataSource)
  },
  // Union field data can be only one of the following:
  "steps": {
    object (Steps)
  },
  "floors": {
    object (Floors)
  },
  "heartRate": {
    object (HeartRate)
  },
  "sleep": {
    object (Sleep)
  },
  "dailyRestingHeartRate": {
    object (DailyRestingHeartRate)
  },
  "dailyHeartRateVariability": {
    object (DailyHeartRateVariability)
  },
  "exercise": {
    object (Exercise)
  },
  "weight": {
    object (Weight)
  },
  "altitude": {
    object (Altitude)
  },
}
```

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Resource: DataPoint

Steps

ObservationTimeInterval

CivilDateTime

Date

TimeOfDay

Floors

HeartRate

ObservationSampleTime

HeartRate.HeartRateMetadata

HeartRate.HeartRateMetadata.MotionContext

HeartRate.HeartRateMetadata.SensorLocation

Sleep

SessionTimeInterval

Sleep.SleepType

Sleep.SleepStage

Sleep.SleepStageType

Sleep.OutOfBedSegment

Sleep.SleepMetadata

Sleep.SleepMetadata.StagesState

Sleep.SleepSummary

Sleep.SleepSummary.StageSummary

DailyRestingHeartRate

DailyRestingHeartRate.DailyRestingHeartRateMetadata

DailyRestingHeartRate.DailyRestingHeartRateMetadata.CalculationMethod

DailyHeartRateVariability

Exercise

Health **Privacy** Is Always Critical

Protected Health Information (PHI): any individually identifiable health data held or sent by a covered entity or its business associates, covering past, present, or future physical or mental health conditions, treatment, and payment details.

Example: Social Security, medical record, health plan beneficiary numbers, billing data

- Under US federal law, PHI is strictly safeguarded by the **HIPAA** (Health Insurance Portability and Accountability Act) Privacy Rule.
- Consumer health apps may sit outside that boundary. Each company's privacy policy and terms of service define what it does with user data
- Map other applicable laws, contracts, and institutional policies

Governance: Systems for Intended Use

Health governance assigns accountable roles, policies, and technical controls across a system's lifecycle.

Governance turns a system's intended use into operating controls:

- **Purpose and access:** function, users, and permitted roles
- **Data and output quality:** validation, provenance, and review
- **Lifecycle:** retention, deletion, monitoring, change, and corrective action

Contains Nonbinding Recommendations

Clinical Decision Support Software

Guidance for Industry and Food and Drug Administration Staff

Document issued on January 29, 2026.

This document supersedes “Clinical Decision Support Software” issued on January 6, 2026.

For questions about this document regarding CDRH-regulated devices, contact the Digital Health Center of Excellence via email at DigitalHealth@fda.hhs.gov. For questions about this document regarding CBER-regulated devices, contact the Office of Communication, Outreach, and Development (OCOD) at 1-800-835-4709 or 240-402-8010, or by email at industry.biologics@fda.hhs.gov. For questions about this document regarding CDER-regulated products, contact Center for Drug Evaluation and Research, Food and Drug Administration at druginfo@fda.hhs.gov. For questions about this document regarding combination products, contact the Office of Combination Products at combination@fda.gov.



U.S. Department of Health and Human Services
Food and Drug Administration
Center for Devices and Radiological Health
Center for Biologics Evaluation and Research
Center for Drug Evaluation and Research
Office of Combination Products in the Office of the Commissioner

Lab: Colab, Health-Data Streams, and a First LLM API Call

Practice the tools we will use throughout the course with small synthetic examples.

Lab Overview

1. **Colab basics** — run cells and understand runtime state
2. **Synthetic health data** — explore steps, heart rate, and GPS
3. **LLM API demos** — prompts of calling an LLM and a step-pattern summary
4. **Take-home** — plot one stream; explain what it can and cannot reveal

Complete Without Devices or Credentials

- The complete lab uses deterministic synthetic data generated in the notebook.
- No device, personal health data, download, map service, or GPU is required.
- If an `OPENAI_API_KEY` is available in Colab Secrets, you may run the three small API demonstrations.
- Without a key, those calls skip cleanly; the data exercises and take-home remain complete.
- Devices and durable class API keys begin in **Week 03**.

Already comfortable with Colab? Skim Part 1 and begin with the synthetic-data exercises in Part 2.

Thanks for the Attention
Q&A